Polyconvexity, Invariants, and Why We Built Ours This Way

A didactic companion to the anisotropic PANN construction

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Abstract

This is a teaching document, not a paper. Its only goal is for you to *really* understand, from the ground up, what polyconvexity is, why hyperelasticity needs this specific and slightly strange-looking condition instead of ordinary convexity, what the "three famous invariants" I_1, I_2, I_3 actually are and why they are polyconvex, and – the part that is genuinely specific to this project – what our own 15 features are, why they look the way they do, whether they are a generalization of I_1, I_2, I_3 or something else entirely, and where the idea came from. Every numerical example below was computed and checked, not invented, so you can reproduce every number yourself.

Contents

1 Why do we care about convexity at all?

Start from the most basic possible question, with no elasticity in it yet: why is convexity such a big deal in optimization and in physics?

> *A convex function has no "fake" local minima to get stuck in, and a line connecting any two points on its graph always stays above the graph. This one geometric fact is what makes minimization problems built from convex functions well-behaved: gradient-based solvers (including Newton's method) converge reliably, and a minimizer is guaranteed to exist under mild extra conditions.*

Take the simplest possible example, $f(x) = x^2$. It has one minimum, at $x = 0$, and Newton's method finds it in one step from anywhere. Now compare $g(x) = x^4 - 3x^2$ (a double-well shape): it has *two* local minima and a local maximum in between. A gradient-based solver started near the "wrong" well will happily converge to a minimizer that is not the global one, and if you are minimizing an energy that is supposed to represent physical equilibrium, converging to the wrong local minimum is a real, physical wrong answer, not just a numerical inconvenience.

In hyperelasticity, the "optimization problem" is: find the displacement field that minimizes the total elastic energy, subject to the boundary conditions. If the energy density W (as a function of the deformation) is convex, classical results (the *direct method of the calculus of variations*) give you existence of a minimizer more or less for free, and Newton's method on the discretized problem tends to behave itself. This is the entire motivation for wanting some kind of convexity. The question is: convexity of *what*, exactly?

2 The naive answer fails: W cannot be convex in F

The most naive thing to ask for is that W, as a function of the deformation gradient F (a 2×2 matrix, in our 2D setting), be convex in F. This fails, and it fails for a clean, checkable reason that has nothing to do with any specific material: *objectivity* (frame indifference) makes it impossible.

Objectivity says $W(QF) = W(F)$ for every rotation $Q \in \mathrm{SO}(2)$: rotating the deformed body rigidly, without straining it further, cannot change the stored energy. Take $Q = -I$ (a 180° rotation, which is a genuine element of $\mathrm{SO}(2)$ in two dimensions). Objectivity forces

$$W(-F) = W(F).$$

Now if W were convex in F, the value at the midpoint of the segment from F to $-F$ would have to be no larger than the average of the endpoints:

$$W\big(\tfrac{1}{2}F + \tfrac{1}{2}(-F)\big) = W(\mathbf{0}) \;\le\; \tfrac{1}{2}W(F) + \tfrac{1}{2}W(-F) = W(F).$$

But the midpoint is the *zero matrix*, $F = \mathbf{0}$, which has $\det F = 0$: physically, this is a state where the material has been crushed to zero volume, and any physically sensible energy must blow up there ($W \to \infty$), not stay finite and below $W(F)$. Convexity in F and objectivity are flatly incompatible. This is not a defect of any particular model; it is a theorem (it goes back to Coleman and Noll, and is discussed in essentially every modern treatment, e.g. Ciarlet's textbook [2]).

> *You cannot require W to be an ordinary convex function of F. Any usable notion of convexity for elasticity has to be something weaker, compatible with $W(-F) = W(F)$ and with $W \to \infty$ as the material is crushed.*

3 Ball's trick: lift the problem into a bigger space

Here is the central idea, and it is worth understanding it once, in the abstract, completely divorced from elasticity, because then the elasticity version stops looking mysterious.

> *A function can fail to be convex in its* original *variable, and yet become convex the moment you feed it some* extra, derived *quantity as if that quantity were an independent input.*

Worked example 1 (The lifting trick, with no elasticity in it)**.** *Let $F = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ and consider the function $h(F) = \det F = ad - bc$. As a function of the four numbers (a, b, c, d), h is a saddle-shaped bilinear form: its Hessian is indefinite (it has both positive and negative eigenvalues), so*

h is not *convex in F. Nudging b and c in the right directions can decrease h below the average of two points where it is large – exactly what convexity forbids.*

Now play a trick: introduce a new, extra symbol δ, declare $\delta := \det F$, and consider the function of the pair *(F, δ) given by*

$$\tilde{h}(F, \delta) := \delta.$$

This is now trivially convex (it is even affine/linear) in the pair (F, δ), because it does not depend on F at all once δ is treated as a free-standing coordinate! Of course $\tilde{h}(F, \det F) = h(F)$ recovers the original, non-convex function – nothing physical changed. What changed is that we are now allowed to ask a different, easier *question: is $\tilde{h}$ jointly convex in the enlarged pair of independent variables (F, δ), rather than in F alone along the curved, constrained set $\{(F, \det F)\}$? The answer can be yes even when the original question's answer is no.*

This is, almost verbatim, what Ball [1] does for elasticity. Instead of asking whether W is convex in F, he asks whether there exists a function G of *three independent block variables* $(F, \operatorname{cof} F, \det F)$ – treating the cofactor matrix $\operatorname{cof} F$ and the determinant $\det F$ as if they were extra, free-standing inputs, exactly like δ above – such that

$$W(F) = G(F, \operatorname{cof} F, \det F), \qquad G \text{ jointly convex on } \mathbb{R}^{2\times 2} \times \mathbb{R}^{2\times 2} \times \mathbb{R}. \tag{1}$$

This is *polyconvexity.* It is strictly weaker than convexity in F (every F-convex function is trivially polyconvex, by ignoring the extra blocks, but not conversely), it is compatible with objectivity (because $\operatorname{cof}(-F) = \operatorname{cof} F$ and $\det(-F) = \det F$ in 2D, so $G(-F, \operatorname{cof} F, \det F)$ need not equal $G(F, \operatorname{cof} F, \det F)$ for the same reason δ didn't care about the sign flip trick above – the non-convex, sign-sensitive part of the problem is exactly the part that got "absorbed" into the extra blocks), and it is exactly the condition Ball needed, together with a growth condition, to prove existence of energy minimizers by the direct method. It says nothing, by itself, about the sign of $\partial^2 W/\partial F^2$ at a given deformation – that is a completely different, and strictly stronger, question, discussed in the main paper.

4 The three famous invariants

Now to your direct question: yes, you have heard of three famous invariants, and here they are. For the right Cauchy–Green tensor $C = F^T F$ (symmetric, positive definite), the classical invariants used in essentially every hyperelasticity textbook (see Ogden [3], or Ciarlet [2]) are

$$I_1 = \operatorname{tr} C, \tag{2}$$

$$I_2 = \tfrac{1}{2}\big[(\operatorname{tr} C)^2 - \operatorname{tr}(C^2)\big] = \operatorname{tr}\big(\operatorname{cof}(C)\big) \quad \text{(in 3D)}, \tag{3}$$

$$I_3 = \det C = J^2. \tag{4}$$

Physically: I_1 is (twice) the sum of the squared principal stretches, so it measures how much the material has been stretched overall, in all directions combined; $I_3 = J^2$ measures the squared change in volume; I_2 is related to how much *areas* (rather than lengths) have changed, which is why it is built from $\operatorname{cof}(C)$, the object that maps how oriented area elements transform under F.

The two most famous compressible/incompressible hyperelastic models in existence are built directly from these:

$$W_{\text{neo-Hookean}} = c_1(I_1 - 3), \qquad W_{\text{Mooney-Rivlin}} = c_1(I_1 - 3) + c_2(I_2 - 3),$$

and both are polyconvex, for a clean structural reason you can already see: $I_1 = \operatorname{tr} C = |F|^2$ (a sum of squares of the entries of F) is convex in F directly – it lives entirely in the *first* block of (1) – and I_2, being built from $\operatorname{cof}(C)$, lives in the *second* block. For non-negative coefficients $c_1, c_2 \geq 0$, a non-negative combination of functions that are each convex in their own block is again jointly convex in the combined variable $(F, \operatorname{cof} F, J)$ – exactly the "sum of convex pieces is convex" rule used throughout the main paper.

Fact 1 (A 2D subtlety worth knowing). *In three dimensions, I_1, I_2, I_3 are three genuinely independent numbers. In two dimensions – our setting – C has only two eigenvalues λ_1, λ_2 (squared principal stretches), and the "I_2" role (sum of products of pairs of eigenvalues) collapses onto a single pair, $\lambda_1\lambda_2 = \det C = I_3$. So in 2D, the classical I_2 and I_3 are literally the same number, and the naive "three invariants" picture only has two independent scalars, I_1 and J. This is exactly why Ball's 2D theory is phrased using the whole* matrix $\operatorname{cof} F$ *as a block, not just its scalar trace: treating* $\operatorname{cof} F$ *as a full 2×2 object (not merely "the second invariant") leaves room to extract genuinely new, non-isotropic information from it, which a single scalar trace could never do (its trace, as just noted, only gives you I_1 back). This is precisely the room our own construction below uses.*

5 Are we using I_1, I_2, I_3? – Yes, generalized

Here is the direct answer to "are we using the three famous invariants," made precise rather than hand-wavy.

Fact 2 (I_1 is a very special case of our own construction). *A basic, exact identity: for* any *orthonormal basis $\{e_1, e_2\}$ of $\mathbb{R}^2$ (not just the standard one),*

$$I_1 = \operatorname{tr} C = e_1^T C e_1 + e_2^T C e_2.$$

That is, I_1 is already*, secretly, a weighted sum of $d \cdot Cd$ over two chosen unit directions, with weight 1 on each and power $p = 1$. Compare this to our own feature family (main paper, Eq. for $Q^F_{k,p}$),*

$$Q^F_{k,p} = \sum_{i=1}^{3} w_{ki} \, (d_{ki} \cdot C d_{ki})^p.$$

I_1 is exactly the special case of this formula with: only two *directions instead of three, those two directions* orthogonal *to each other, weights* equal to 1*, and power $p = 1$. Our construction is not analogous to I_1; it is a literal, direct generalization of it – more directions (three, not two), no requirement that they be orthogonal, general positive weights (subject to the balance condition below, which is exactly what replaces "orthonormal" once you give up orthogonality), and higher powers $p = 2, 3$ instead of $p = 1$.*

The same story holds one level up: our $Q^H_{k,p}$ features use $d \cdot \operatorname{cof}(C)d$ in exactly the same way Q^F uses $d \cdot Cd$, so they generalize the cof-based (I_2-flavoured) invariant in exactly the same sense. And J, J^2 are not generalized at all – they are I_3 and I_3^2, used completely unchanged, because the volumetric invariant does not need an anisotropic generalization: volume change is already a single, direction-independent scalar, and there is nothing anisotropic to generalize.

> *So: yes, we are using I_1, I_2 (via* cof*), and $I_3 = J^2$ – but not as three bare numbers. We use* many *directional versions of the I_1-type and I_2-type constructions (three direction systems, each contributing terms at powers $p = 2, 3$), because a material with* no assumed symmetry *needs to be able to respond differently in different directions, and a single scalar I_1 or I_2, being built from an orthonormal (or in fact* any*) basis by summation, is completely blind to direction by construction.*

6 Why I_1, I_2, I_3 alone cannot describe an anisotropic material

This needs to be seen numerically to really land, so here it is, computed for real, not asserted.

Worked example 2 (Two different materials that I_1 and J cannot tell apart)**.** *Take two deformation states,*

$$F_a = \left(\begin{smallmatrix} 1.4 & 0 \\ 0 & 0.9 \end{smallmatrix}\right) \quad (\textit{stretch } 1.4\times \textit{ along } x,\ \textit{compress to } 0.9\times \textit{ along } y), \qquad F_b = \left(\begin{smallmatrix} 0.9 & 0 \\ 0 & 1.4 \end{smallmatrix}\right) \quad (\textit{roles of } x, y \textit{ swapped}).$$

These are physically very different states for an anisotropic material – one stretches hard along x and compresses along y, the other does the opposite – and a material whose stiffness depends on direction should, in general, store a different amount of energy in each. Yet:

$$I_1(F_a) = I_1(F_b) = 2.77, \qquad J(F_a) = J(F_b) = 1.26.$$

Exactly equal, to every decimal place, because trace and determinant literally do not know which axis is which. Any *energy built purely from I_1, I_2, I_3 (i.e., any* isotropic *hyperelastic model, no matter how the invariants are combined) is mathematically incapable of distinguishing F_a from F_b. This is not a approximation error; it is exact and structural.*

7 Our construction: direction systems, balance, and why the cubic power is not decorative

To fix Example 2, you need at least one feature that is *not* blind to direction. That is what the three direction systems (Table of direction systems in the main paper) are for. Each system k consists of three unit vectors d_{k1}, d_{k2}, d_{k3} at specific angles, with specific positive weights w_{k1}, w_{k2}, w_{k3}, satisfying the *balance condition*

$$\sum_{i=1}^{3} w_{ki}\, d_{ki} \otimes d_{ki} = I. \tag{5}$$

This condition is the direct 2D generalization of "orthonormal with unit weights": it says the chosen (possibly oblique, non-orthogonal) directions, weighted appropriately, still add up to the identity, the same way $e_1 \otimes e_1 + e_2 \otimes e_2 = I$ does for an orthonormal pair. This is exactly what makes the reference-state stress-free proof (C4 in the main paper) go through with a single scalar correction, and it is why the model behaves *isotropically at the undeformed state* even though it is fully anisotropic away from it – a real material with a texture or weave usually looks locally isotropic right at rest and only reveals its anisotropy once you deform it, which is exactly the behavior this construction reproduces by design.

Worked example 3 (Computing $Q^F_{1,2}$ and $Q^F_{1,3}$ by hand, for real)**.** *System 1 uses angles* $0°, 50°, 130°$ *with weights* $0.2959, 0.8520, 0.8520$. *For F_a above, $C = \mathrm{diag}(1.96, 0.81)$. The three directions are $d_1 = (1, 0)$, $d_2 = (0.6428, 0.7660)$, $d_3 = (-0.6428, 0.7660)$, giving*

$$d_1 \cdot Cd_1 = 1.96, \qquad d_2 \cdot Cd_2 = d_3 \cdot Cd_3 = 1.2852.$$

So

$$Q^F_{1,2}(F_a) = 0.2959(1.96)^2 + 2 \times 0.8520(1.2852)^2 = 3.9511,$$

$$Q^F_{1,3}(F_a) = 0.2959(1.96)^3 + 2 \times 0.8520(1.2852)^3 = 5.8449.$$

Repeating for F_b (swap the roles of x, y, so effectively $d \cdot Cd$ picks up the 0.81 *and* 1.96 *in the opposite pattern) gives*

$$Q^F_{1,2}(F_b) = 3.95107, \qquad Q^F_{1,3}(F_b) = 5.7357.$$

Now compare the two states directly:

$$\left|Q^F_{1,2}(F_a) - Q^F_{1,2}(F_b)\right| \approx 0.00001, \qquad \left|Q^F_{1,3}(F_a) - Q^F_{1,3}(F_b)\right| \approx 0.109.$$

This is the single most important number in this whole document. ***The power-2 feature barely tells*** F_a ***and*** F_b ***apart*** *(a difference in the fifth decimal, for this system) – it turns out a balanced* quadratic *aggregate in 2D obeys an almost-exact hidden symmetry under swapping the two axes, so it is a nearly-blind, weak anisotropy detector, not so different from* I_1 *itself in this respect.* ***The power-3 feature, by contrast, differs by about*** 1.9% ***of its own value*** *– a real, structural, non-accidental difference. This is exactly why the model uses both* $p = 2$ *and* $p = 3$ *terms for every direction system, and why the main paper states this is "not cosmetic": without the cubic terms, the model would be far closer to isotropic than intended, almost by accident, for reasons that have nothing to do with training and everything to do with a 2D geometric coincidence at even powers.*

Why *three* separate direction systems, rather than one? One system of three directions already breaks isotropy (Example 3 proves that), but it only samples the material's directional response along three specific angles. A real anisotropic material's stiffness can vary continuously with direction in a complicated way; three independent, differently-angled systems (nine directions in total, each individually balanced) give the model many more independent "probes" of directional stiffness to combine, exactly as using more sensors at more angles gives a richer picture of an unknown directional signal than a single triple of sensors would.

8 Where did this come from – did we invent it?

Split honestly into two layers, because the honest answer has two different parts.

The overall strategy (not ours). Building a polyconvex energy as an input-convex neural network (ICNN) applied to hand-picked, individually-polyconvex invariants is Klein et al.'s approach [4], itself sitting on top of Ball's original existence theory [1] and the standard invariant-based hyperelasticity tradition going back to Ogden and earlier [3]. Klein et al. tie their invariants to a *named* material symmetry group (cubic, transversely isotropic, . . .) via a classical structural tensor – a well-established, textbook technique (structural-tensor / integrity-basis representation theory for anisotropic invariants) applied by many authors before them.

The specific mechanism (ours). What is genuinely this project's own construction is the decision *not* to tie the invariants to any named symmetry group at all: instead, use several generic direction systems, individually balanced via (5), with no 90°-rotation closure and no D_4 averaging built in anywhere. This is the specific answer to the open problem Linden et al. [5] flag explicitly in their own paper – that a complete, polyconvexity-compatible invariant set may not exist for a general, unknown symmetry class – and it is what lets this construction claim exact objectivity, an exact stress-free reference state, and exact polyconvexity *without ever assuming or checking* which symmetry class, if any, the material belongs to. The specific numerical angles and weights in the direction-systems table are one valid solution of the (underdetermined) balance condition (5) – there are infinitely many valid choices, and the ones used were not derived from any deeper physical principle beyond satisfying (5) to machine precision and providing three visibly different, non-D_4-related angular patterns; if you want a cleaner or differently-optimized choice, the balance condition is the only real constraint you must preserve.

9 A short recap, in one paragraph

Ordinary convexity in F is impossible for any objective energy (Section 2). Ball's fix, polyconvexity, asks for convexity not in F but in the enlarged, "lifted" triple $(F, \operatorname{cof} F, \det F)$, treated as independent variables (Section 3) – exactly the same trick as promoting $\det F$ to its own free coordinate in Example 1. The classical invariants I_1, I_2, I_3 are the textbook way to build polyconvex energies out of this triple (Section 4), but they are isotropic by construction and provably blind to direction (Example 2). Our own features are a direct, literal generalization of I_1 and the cof-based invariant – more directions, general weights satisfying a balance condition, and crucially both quadratic *and cubic* powers, because the quadratic power alone is a nearly-blind anisotropy detector in 2D (Example 3). The overall ICNN-on-invariants strategy is Klein et al.'s; the specific choice to use generic, symmetry-free direction systems instead of a named structural tensor is this project's own contribution, built to answer an open problem stated explicitly in the literature it responds to.

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